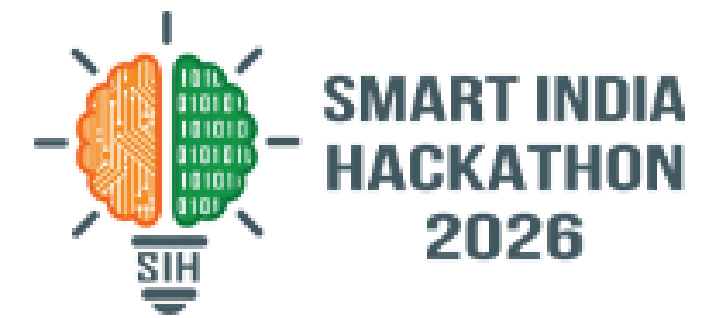


SMART INDIA HACKATHON 2025



- **Problem Statement ID** -26117
- **Problem Statement Title** - Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

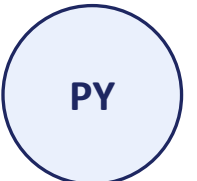
Theme - Smart Automation

- **PS Category** - software
- **Team ID** -
- **Team Name** -
(Registered on portal)



TECHNICAL APPROACH

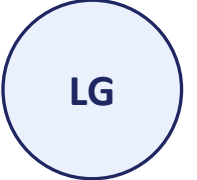
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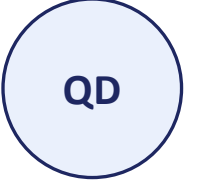
Python / FastAPI



Ollama / vLLM



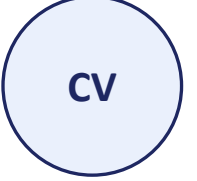
LangGraph Agent



Qdrant (RAG)

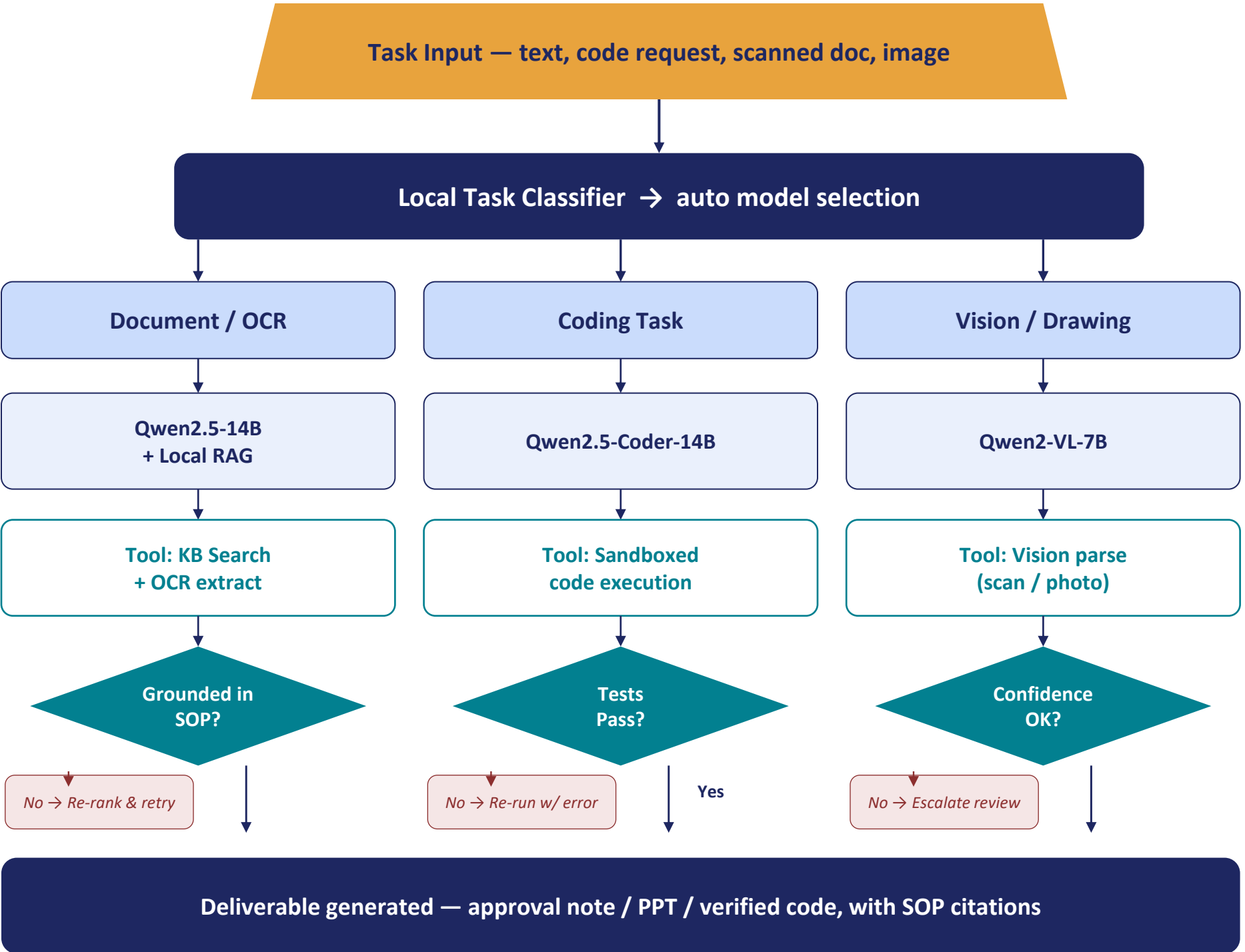


Docker Sandbox



PaddleOCR / VLM

AGENT & MODEL-ROUTING PIPELINE



OPEN-WEIGHT MODELS ON-PREM: Qwen2.5-14B (general) · Qwen2.5-Coder-14B (code) · Qwen2-VL-7B (vision) · BGE-M3 (embeddings) · PaddleOCR

SOVEREIGN AI WORKBENCH

● System secure — 0 external calls detected

"Read this scanned inspection report and draft an approval note."

Routed → Qwen2.5-14B + RAG

Plan: OCR scan → search SOP → draft note → verify
Tool calls: ocr_extract() · search_kb() · generate_docx()

Files generated

- ✓ Approval_Note.docx
- ✓ calc_check.py — verified in sandbox

NETWORK AUDIT MONITOR

Outbound requests today: 0
All model + tool calls served on-premise

STACK: Ollama/vLLM · LangGraph · FastAPI · Qdrant · Docker (no-net sandbox) · iptables + live traffic monitor

FEASIBILITY AND VIABILITY

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100% AIR-GAPPED

✓ Feasibility of idea

- Strong, proven demand — PSUs, refineries and defence units already handle this work manually
- Open-weight multimodal models (Qwen, Llama, DeepSeek) now perform close to commercial-grade quality
- Mature open-source stack — Ollama/vLLM, LangGraph, Qdrant — lets a working prototype ship fast
- Config-driven router lets new open-weight models slot in without re-architecting the system

! Challenges and risks

- On-prem GPU budget caps which model sizes can be served at once
- Open-weight models can still lag frontier cloud models on complex, multi-step reasoning
- OCR/vision accuracy drops on poor scans, handwriting and low-quality drawings
- The open-weight landscape moves fast — a fixed integration can go stale within months

→ Strategies to overcome

- Start with 7B–14B quantized models sized to available hardware; scale as GPU budget grows
- Route every task to the model best suited for it, instead of one generalist model
- Keep a human review gate before any approval note or deliverable is finalized
- Run all code execution in a network-isolated sandbox; encrypt data at rest
- Ground every output in the organization's SOPs via RAG, with citations for audit
- Config-driven router lets new open-weight models be added without redesign

IMPACT & BENEFITS

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2026**

POTENTIAL IMPACTS



Eliminates risk of confidential data ever leaving refinery premises



Retrieves instant answers from SOPs, manuals & past correspondence



Auto-drafts approval notes, Word/PPT reports and calculations



Runs and verifies AI-generated code safely in a sandbox



Cuts manual review time for scanned inspection reports & drawings



Scales to new open-weight models without redesigning the system



Operational Benefit

- ☐ Frees engineers from repetitive documentation work
- ☐ Speeds up approval cycles for maintenance & inspection
- ☐ Keeps consistent, audit-ready records automatically



Economic Benefit

- ☐ Removes recurring cloud-AI subscription & API costs
- ☐ Reuses existing on-premise GPU infrastructure
- ☐ Cuts man-hours spent on manual document review

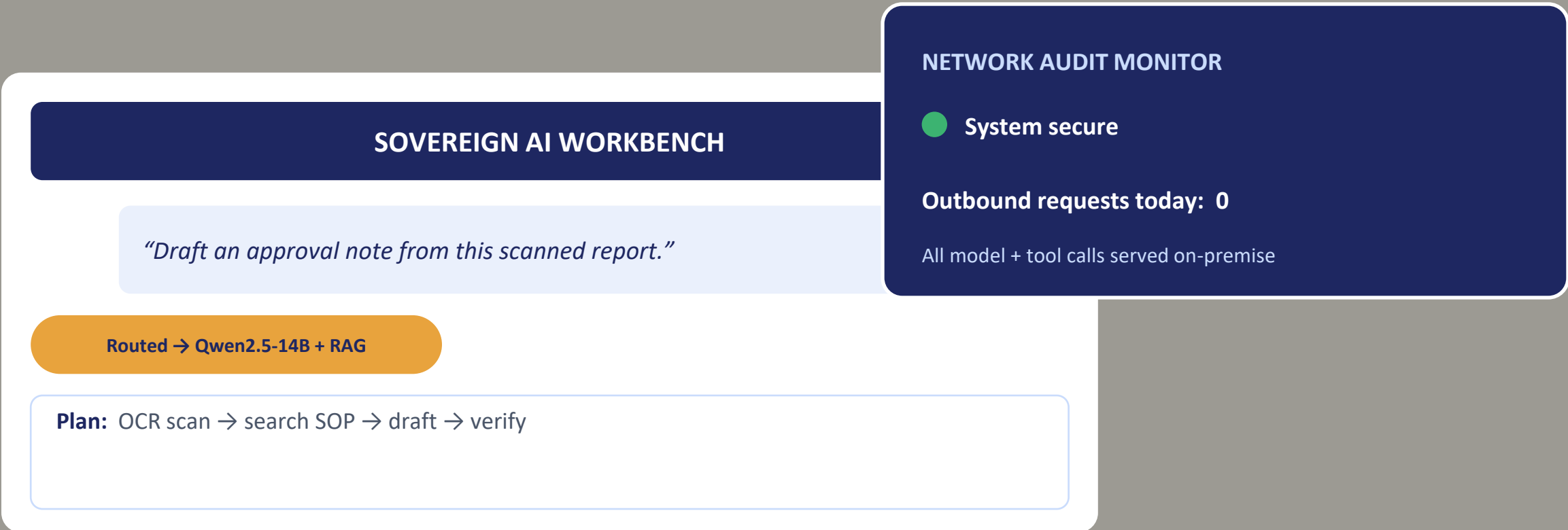
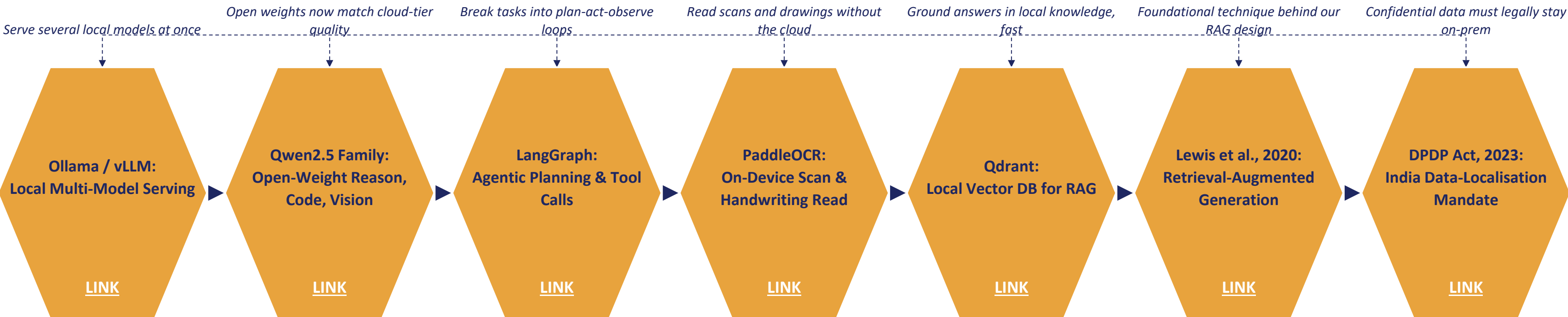


Strategic & Security Benefit

- ☐ Zero data exfiltration — meets defence-grade confidentiality norms
- ☐ Aligns with Atmanirbhar Bharat / Make-in-India objectives
- ☐ Future-proof: new open-weight models added without re-engineering

RESEARCH AND REFERENCES

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UI / UX

Illustrative UI concept for the on-prem workbench console — not a live product screenshot